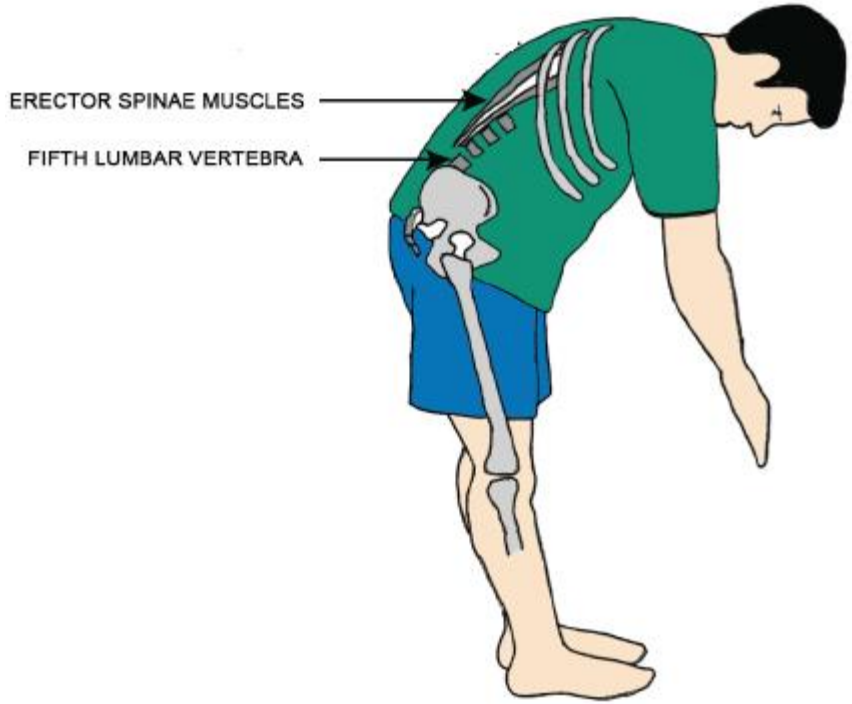
Static analysis – spine





Case I
$\theta = 85^\circ$

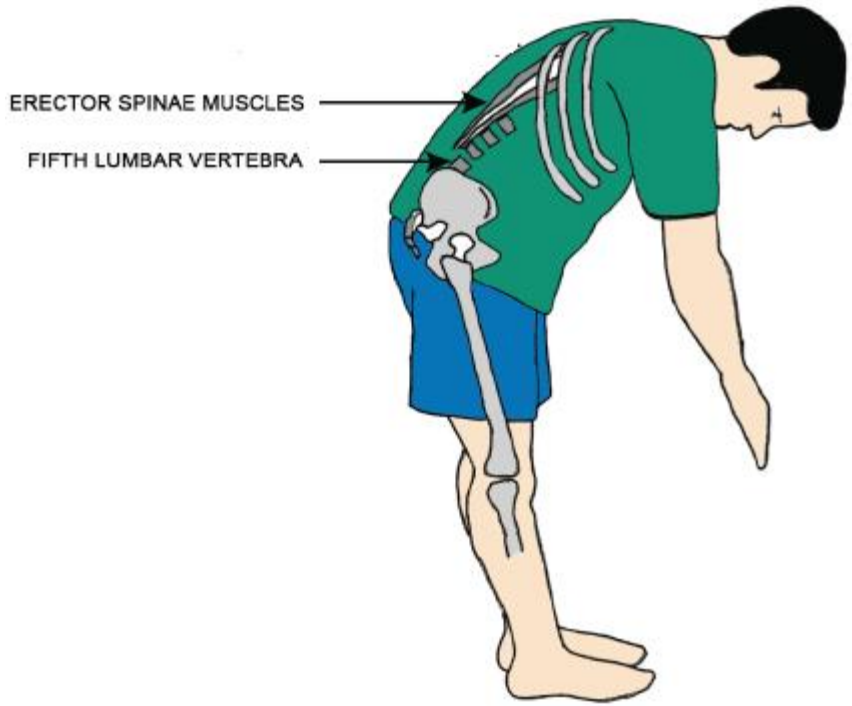


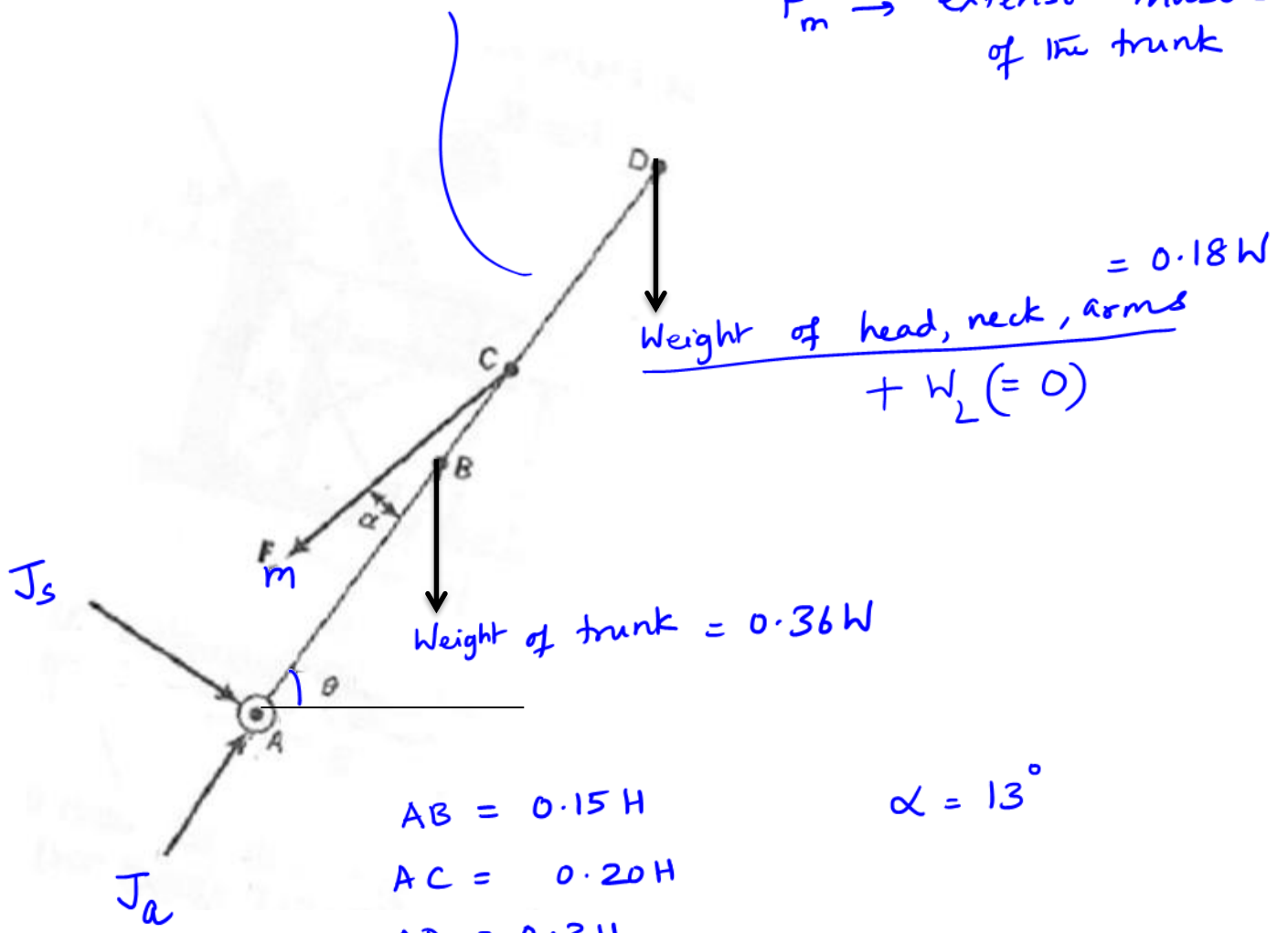
Case II
$\theta = 75^\circ$



Case III
$\theta = 60^\circ$

Static analysis – spine





Static analysis – spine

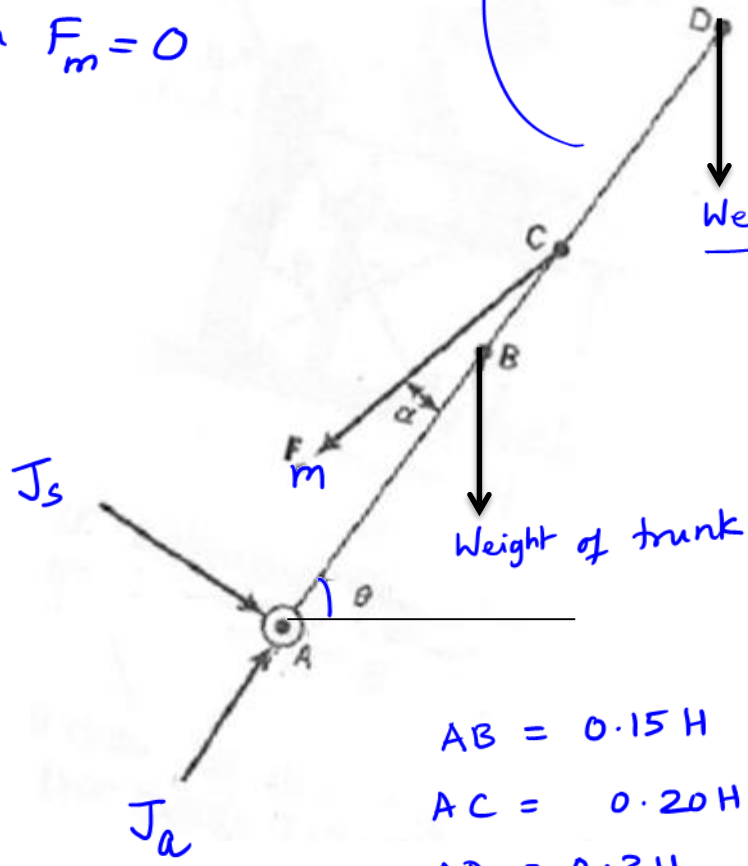
Person is carrying $W_L = 0.2W$ equally distributed into a 0.1W load in each hand. Use the FBD to analytically determine θ at which $F_m = 0$



Thoracolumbar spine

A → lumbosacral joint

F_m → extensor muscles of the trunk



Weight of head, neck, arms + W_L = 0.18W + 0.2W

Weight of trunk = 0.36W

AB = 0.15 H

AC = 0.20H

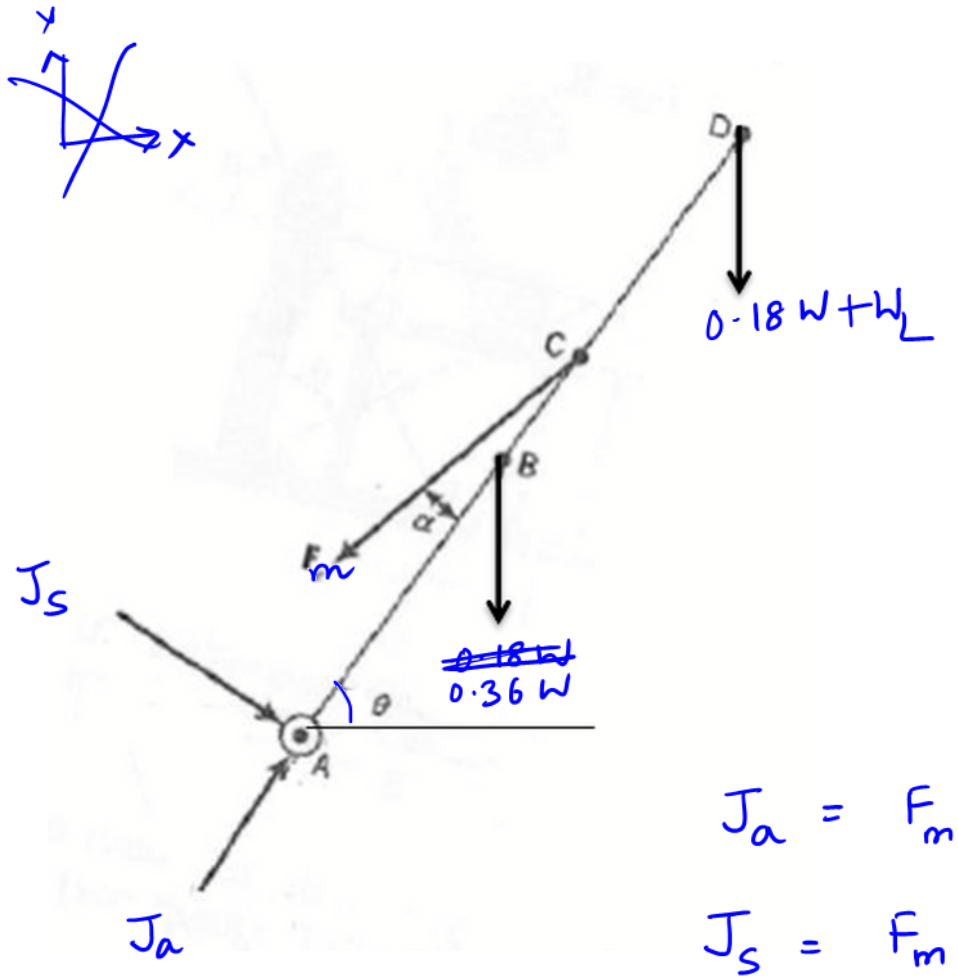
AD = 0.3H

$\alpha = 13^\circ$

$$F_m = \frac{0.36W(AB)\cos\theta + (0.18W + W_L)\,AD\cos\theta}{AC\sin\alpha}$$

$J_a = 0.74W$

$J_s = 0$



$\Sigma M_A = 0$, $\Sigma F_x = 0$, $\Sigma F_y = 0$

$\circlearrowleft +$

$-(0.36W)(0.15H)\cos\theta - (0.18W)\,0.3H\cos\theta + F_m \sin\alpha\,(0.2H) = 0$

$$F_m = \frac{0.108\,W/\!\!/ \cos\theta}{0.2/\!\!/ \sin\alpha}$$

for non-zero W_L

$$F_m = \frac{(0.108W + 0.3W_L)\cos\theta}{0.2\sin\alpha}$$

$J_a = F_m \cos\alpha + 0.36W\sin\theta + 0.18W\sin\theta$

$J_s = F_m \sin\alpha - 0.36W\cos\theta - 0.18W\cos\theta$

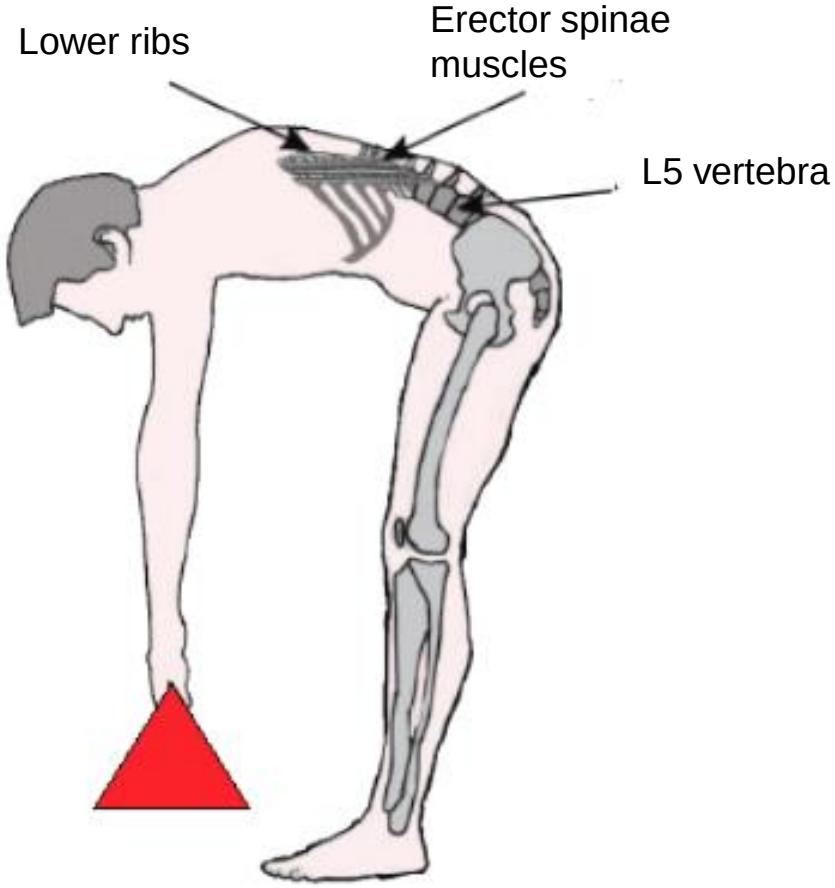
	$\theta = 85°$	$\theta = 75°$	$\theta = 60°$
F_m	$0.209\,W$	$0.62\,W$	$1.2\,W$
J_a	$0.74\,W$	$1.13\,W$	$1.64\,W$
J_s	~ 0		

Ergonomics or Human factors

Static analysis – spine

for $W_L = 0.2\,W$

	$\theta = 35^\circ$	$\theta = 75^\circ$	$\theta = 85^\circ$
F_m	$1.89\,W$	$0.97\,W$	$0.32\,W$
J_a	$3.4\,W$	$1.7\,W$	$1.05\,W$
J_s	$0.08\,W$	$0.024\,W$	$0.008\,W$





Lifting with the back

$\theta = 35^\circ$



Lifting with the legs

$\theta = 75^\circ$



Dividing the load and lifting with handles

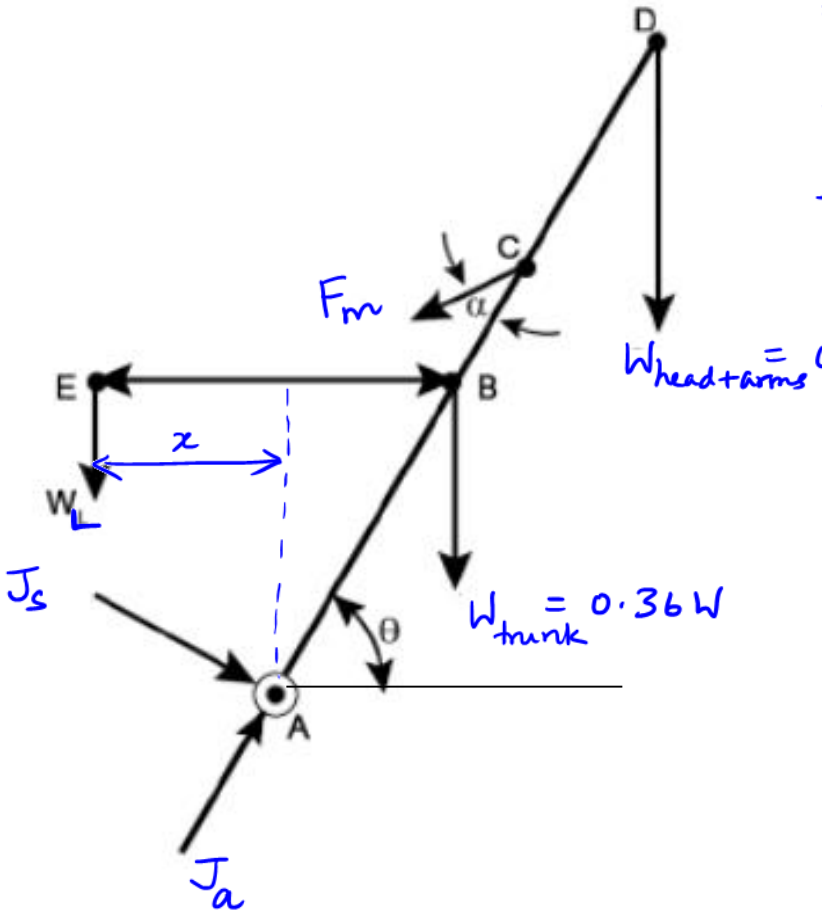
$\theta = 85^\circ$

Static analysis – spine

Determine the angle θ at which $F_m = 0$



$W_L = 0.2W$



Given EB = 0.12H

$x = 0.12H - 0.15H\cos\theta$

$\Sigma M_A = 0$

For $F_m = 0$

$-0.18W\ AD\cos\theta - 0.36W\ AB\cos\theta + W_L(0.12H - 0.15H\cos\theta) = 0$

$\theta = 80^\circ$

$J_a = 0.74W$

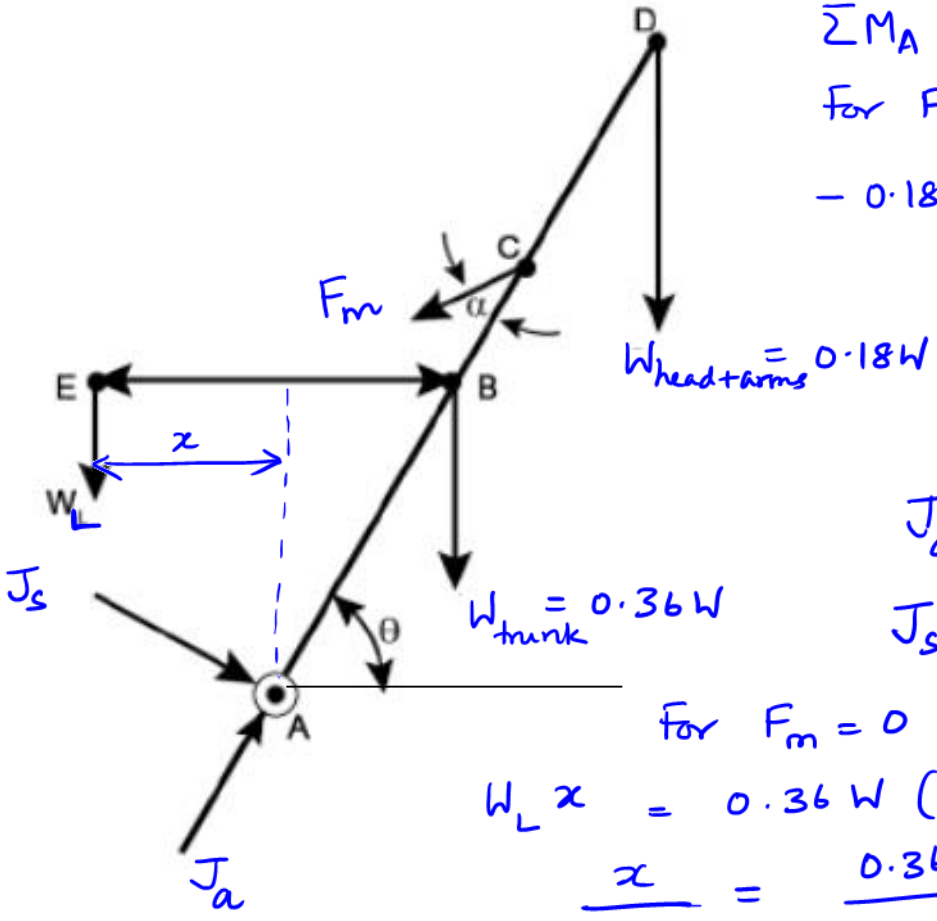
$J_s = 0.13W$

Static analysis – spine

Determine the angle θ at which $F_m = 0$

$W_L = 0.2W$





$x = 0.12H - 0.15H\cos\theta$

$\Sigma M_A = 0$

For $F_m = 0$

$-0.18W\ AD\cos\theta - 0.36W\ AB\cos\theta + W_L(0.12H - 0.15H\cos\theta) = 0$

$\theta = 80^\circ$

$J_a = 0.74W$

$J_s = 0.13W$

For $F_m = 0$

$W_L x = 0.36W(AB\cos\theta) + 0.18W(AD\cos\theta)$

$\frac{x}{\cos\theta} = \frac{0.36W(AB) + 0.18W(AD)}{W_L} = \frac{k\ (\text{const})}{W_L}$

Given $EB = 0.12H$